

DanReality

Dance to Deeper Insights and Growth

RESEARCH THESIS PROPOSAL

Affective Kinematic Synchronization:

Preserving and Innovating Hunza Dance Heritage through
Real-Time Computer Vision for Mental Wellness

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Abstract

DanReality presents "**Affective Kinematic Synchronization**" (**AKS**)—an interdisciplinary research initiative developed by JanBroz, the creative laboratory of twin brothers Sabir and Shakir Jan from Gulmit, Hunza Valley, Pakistan. This proposal bridges *traditional Hunza dance heritage*, *real-time computer vision*, and *Dance Movement Therapy (DMT)* to address the escalating youth mental health crisis across South and Central Asia.

The AKS system leverages **MediaPipe BlazePose** for real-time body, hand, and face tracking to capture and analyze traditional Hunza dance movements, twin synchronization patterns, and emotional expression. Integrated with **Unity 3D**, **React/Node.js**, **Three.js**, and **Blender**, the platform provides immediate biofeedback on posture, rhythm, coordination, and affective state, generating immersive, culturally-rooted visuals for therapeutic engagement.

Key Finding: Meta-analyses demonstrate that Dance Movement Therapy yields medium effect sizes ($d = 0.60$) for psychological outcomes. Dancers exhibit enhanced functional connectivity within the extended mirror neuron system (eMNS), confirming the neuroscientific basis for dance as therapy.

This study proposes a mixed-methods investigation combining qualitative cultural analysis with quantitative movement data and clinical assessment via **PHQ-9** and **GAD-7** scales. The research addresses a critical gap: no existing system integrates culturally-specific traditional dance from Pakistan's northern regions with real-time affective computing for therapeutic purposes.

1. Introduction and Background

1.1 The Founders: JanBroz — Born from the Mountains

Source: JanBroz Portfolio — saber7jan.github.io/JanBroz | Instagram: @janbroz78

The JanBroz artistic practice emerges from lived experience rooted in the rhythms of **Hunza Hareep**—the traditional melodic-rhythmic form played on drums and flutes in Pakistan's Gilgit-Baltistan region. Sabir and Shakir Jan, twin brothers from **Gulmit, Hunza Valley**, developed extraordinary non-verbal synchronization through shared practice of Hareep-influenced movement, inspired by their grandfather—a graceful folk dancer whose movements were living narratives.

Their self-taught practice fused traditional Hunza dance with freestyle, K-pop, and contemporary choreography, culminating in their representation of Pakistan at the **Jubilee Arts 2018 International Talent Showcase in Lisbon, Portugal**—an achievement marking both personal excellence and cultural diplomacy.

"We didn't just dance — we told the story of our mountains, our grandfather's footsteps, and the resilience of a people who have always expressed truth through movement." — Sabir Jan, JanBroz

1.2 The Global Youth Mental Health Crisis

The World Health Organization identifies mental health conditions as the **leading cause of disability among adolescents aged 10–19 years**. In South Asia, where approximately **45% of the population is under 25**, the convergence of rapid digitization, economic precarity, and limited access to culturally appropriate mental healthcare has created an environment where youth psychological distress frequently goes unaddressed.

Pakistan, with one of the world's largest youth populations, faces particular challenges: the *National Mental Health Programme* estimates that **fewer than 500 qualified psychiatrists** serve a population exceeding 240 million. Stigma surrounding psychological intervention remains pervasive, and existing online platforms offer no culturally relevant therapeutic alternatives.

1.3 The Hunza Dance Tradition

Source: Illustration based on Hunza Valley cultural landscape, Gilgit-Baltistan, Pakistan

The Hunza Valley possesses a rich movement heritage encoded in communal dance practices that function as **embodied storytelling**. The traditional **Hareep** accompanies dances such as **Apiso-Nut**—a collective performance where movement narrates agricultural cycles, seasonal transitions, and communal bonds. The **Burushaski language**, spoken by the Hunza people, carries emotional and philosophical weight in lyrical content addressing themes of resilience, joy, mourning, and spiritual connection.

UNESCO Framework: Traditional dance is identified as a priority domain for intangible cultural heritage preservation. The JanBroz lineage represents the intergenerational transmission of embodied knowledge—from grandfather to twin grandsons—that must be preserved and activated for therapeutic purposes.

2. Literature Review

2.1 Dance Movement Therapy: Evidence and Mechanisms

Source: Dance Movement Therapy Research — ADTA (American Dance Therapy Association)

Dance Movement Therapy (DMT), defined by the American Dance Therapy Association as the psychotherapeutic use of movement to promote emotional, social, cognitive, and physical integration, has accumulated substantial empirical support. A landmark meta-analysis by Koch et al. (2019) synthesizing **41 controlled intervention studies** (N = 2,374) found a **medium overall effect size (d = 0.60, p < 0.001)** for DMT on health-related psychological outcomes.

Notably, DMT produced a consistent effect on affect-related conditions including depression and anxiety (d = 0.30, p < 0.001), while effects on quality of life (d = 0.32), interpersonal skills (d = 0.49), and cognitive skills (d = 0.26) demonstrated clinical significance. **Follow-up data indicated therapeutic gains remained stable or increased over 22 weeks post-intervention.**

Network meta-analysis from China (2025): DMT combined with Chinese dance elements ranked as the most effective intervention for depression (SUCRA = 99.7%), demonstrating that culturally-integrated dance practice significantly enhances therapeutic outcomes. This validates DanReality's approach of centring Hunza cultural movement.

2.2 Neuroscience: The Mirror Neuron System and Dance

Source: Frontiers in Human Neuroscience, 2023 | Embodied Cognition in Dancers

Resting-state fMRI studies reveal that dancers exhibit **enhanced intra- and inter-regional functional connectivity within the extended mirror neuron system (eMNS)**—including the left inferior frontal gyrus, ventral premotor cortex, anterior insula, cerebellum, and basal ganglia—relative to non-dancers. This enhanced connectivity correlates with training intensity and creativity measures.

The mirror neuron system plays a pivotal role in the therapeutic processes underlying DMT. When an individual observes another's movement, mirror neurons activate as if the observer were executing the same action, creating a substrate for **empathic resonance, attunement, and intersubjective connection**. In the JanBroz twin practice, this operates in a uniquely intensified feedback loop where each brother's movement simultaneously serves as observed stimulus and executed action.

2.3 Computer Vision and Pose Estimation

Source: Google Research — MediaPipe BlazePose | research.google/blog

The field of human pose estimation has advanced dramatically through deep learning. **MediaPipe BlazePose**, developed by Google, provides real-time tracking of **33 body keypoints** with three degrees of freedom each, achieving near-human annotation accuracy (PCK@0.2 = 97.2%). The system offers Lite and Full model variants optimizing the speed-accuracy tradeoff, making it suitable for deployment across devices from mobile phones to workstations.

Studies integrating MediaPipe with Unity for real-time avatar animation have achieved **frame rates of 55–60 FPS with latency under 50ms** on standard hardware, confirming technical feasibility for immersive therapeutic applications. The integration of spatial and temporal

processing techniques further improves tracking robustness under occlusion and rapid movement conditions typical in Hunza Hareep dance.

2.4 Affective Computing and Emotion Recognition

Source: Affective Computing Research — arXiv:2407.09913 | Emotion Detection through Body Gesture and Face

Affective computing research has established that **body gestures convey emotional information that may exceed facial expressions in reliability and cultural validity**. Machine learning approaches using OpenPose-extracted keypoints combined with ResNet and DenseNet architectures have achieved **emotion classification accuracies exceeding 90%** on benchmark datasets.

The multi-view emotional expressions dataset (MEED) demonstrated that *2D pose estimation effectively captures emotional shifts* across anger, disgust, fear, happiness, sadness, and surprise from multiple viewing angles. These findings directly support AKS's affective feedback layer, which interprets the emotional content encoded in Hunza dance movements and generates responsive visual environments.

2.5 Interpersonal Synchronization: The Twin Advantage

Source: Current Biology, 2024 | Behavior Research Methods, 2024 — Dance Synchrony Studies

Research on interpersonal synchrony in dance reveals that **"bounce synchronization"**—the vertical movement pattern—functions as a supramodal rhythm regulator enabling both music-driven and partner-driven synchrony. Cross-wavelet transform analysis can quantify this coordination in real time, producing **synchrony indices** that drive visual feedback rewarding coordinated movement.

JanBroz's twin practice uniquely embodies this phenomenon. Sabir and Shakir Jan demonstrate naturally developed bounce synchronization through decades of shared Hareep-based practice—making them ideal research subjects for validating the AKS synchronization layer and developing norms for twin therapeutic dyads.

2.6 Digital Preservation of Intangible Cultural Heritage

Source: UNESCO 2003 Convention for the Safeguarding of Intangible Cultural Heritage

UNESCO's 2003 Convention identifies traditional dance as a priority domain for preservation, emphasizing documentation, research, transmission, and protection. The *i-Treasures project* developed an open platform using multi-sensor technology to capture and analyze intangible heritage forms, applying data fusion and probabilistic inference to transform movement data into human-interpretable knowledge.

Contemporary 3D technologies—motion capture, photogrammetry, VR/AR—are increasingly applied to document traditional dances across multiple cultures. The proposed AKS system contributes by developing a **living archive of Hunza dance** that simultaneously serves therapeutic, educational, and cultural preservation functions—ensuring that knowledge transmitted through the JanBroz lineage from grandfather to twin grandsons remains accessible to future generations.

3. Research Objectives and Questions

3.1 Primary Objective

To develop, implement, and validate the Affective Kinematic Synchronization (AKS) system—a real-time computer vision platform that captures, analyzes, and provides biofeedback on traditional Hunza dance movements and twin synchronization patterns for therapeutic application in youth mental health.

3.2 Secondary Objectives

- Document and digitally preserve Hunza dance movement vocabulary through motion capture and 3D asset creation, establishing an open cultural archive rooted in the JanBroz artistic practice.
- Measure the impact of AKS-guided dance sessions on posture quality, rhythmic coordination, emotional expressivity, and interpersonal synchronization.
- Assess changes in mental health metrics (depression, anxiety, wellbeing) among youth participants following a structured 12-week AKS intervention.
- Develop ethically grounded protocols for culturally-sensitive technology deployment in South Asian communities, addressing data privacy and indigenous data sovereignty.

3.3 Research Questions

RQ1: How can MediaPipe-based real-time body tracking be optimized to accurately capture the distinctive movement qualities of Hunza traditional dance, including Hareep rhythm patterns and twin synchronization dynamics?

RQ2: What quantitative movement features (joint angles, velocity profiles, synchronization indices) correlate with emotional expression and therapeutic engagement in AKS-guided sessions?

RQ3: Does participation in an AKS-based DMT intervention produce statistically significant improvements in depression (PHQ-9), anxiety (GAD-7), and quality of life among youth aged 16–30?

RQ4: How can immersive generative visuals (Unity 3D/Three.js) be designed to respond dynamically to dancers' emotional intent while maintaining cultural authenticity?

RQ5: What ethical frameworks ensure that technology-enhanced preservation of Hunza dance heritage respects indigenous knowledge systems and benefits source communities?

4. Methodology

4.1 Research Design

This study employs a **mixed-methods convergent design** combining quantitative experimental assessment with qualitative ethnographic inquiry and technical prototyping. The three methodological strands operate iteratively: qualitative cultural analysis informs technical system design; quantitative movement metrics validate system performance; and clinical outcomes assessment evaluates therapeutic efficacy.

4.2 Qualitative Strand: Cultural Movement Analysis

The qualitative component involves **in-depth phenomenological interviews** with Sabir and Shakir Jan exploring their embodied experience of Hunza dance tradition, their grandfather's influence, twin synchronization, and the emotional narrative structure of their choreographic work. Video analysis of JanBroz performance archives—including documentation from **Jubilee Arts 2018 in Lisbon**—will be conducted using movement analysis frameworks to identify characteristic movement qualities, rhythmic patterns, and synchronisation dynamics.

4.3 Quantitative Strand: Clinical Outcomes

Source: Patient Health Questionnaire (PHQ-9) & Generalized Anxiety Disorder Scale (GAD-7) — Validated Clinical Instruments

Participants: A purposive sample of **60 youth aged 16–30** will be recruited through community organizations in Hunza Valley and Islamabad, stratified by baseline mental health status (mild-moderate depression/anxiety) and dance experience.

Intervention: Participants complete **12 weekly AKS-guided dance sessions** (90 minutes each) structured around Hunza Hareep rhythms, JanBroz choreography, and freestyle movement with real-time biofeedback. Control group (n=30) receives treatment as usual (TAU).

Clinical Assessment: Primary outcomes via **PHQ-9 and GAD-7** at baseline, mid-intervention (week 6), post-intervention (week 12), and 8-week follow-up. Secondary measures include **WEMWBS** (Warwick-Edinburgh Mental Wellbeing Scale) and the Body Awareness Questionnaire.

4.4 Technical Prototype: AKS Architecture

Source: Unity 3D Real-Time Avatar Animation Research | AKS Platform Architecture

The AKS technical architecture integrates five components:

- MediaPipe Holistic — real-time body (33 keypoints), hand (21 per hand), and face (468 landmarks) extraction at 30+ FPS
- Unity 3D — immersive environment rendering with Hunza cultural motifs (mountain landscapes, traditional textile patterns, celestial navigation symbols)
- React + Node.js + Socket.io — real-time web platform delivery and multi-user session management
- Three.js — browser-based generative visual effects responding to movement affective features
- Blender — 3D asset creation including Hunza-inspired environmental motifs and cultural symbols

A Python-based analytics module processes movement streams to extract affective and kinematic features, driving real-time visual feedback through WebSocket communication.

5. SWOT Analysis

The DanReality project carries inherent strengths from the founders' unique position as Hunza-born twin performers with a multidisciplinary team, while acknowledging early-stage challenges.

STRENGTHS • Twin synchronisation expertise unique in DMT research • Full-stack team: therapy, choreography, technology • Cultural authenticity: Hunza Hareep + Burushaski lyrics • Aditi: 8+ years movement therapy experience • Faizan: 5 years in online dance studio platforms	WEAKNESSES • First-time multidisciplinary team collaboration • Limited initial funding for world-class choreographers • No prior AKS deployment in clinical settings • Hardware requirements for low-resource settings
OPPORTUNITIES • No competitor integrates storytelling + dance therapy + AI • Pakistan youth mental health crisis creates urgent need • UNESCO-aligned cultural preservation mandate • Online dance market projected to reach \$5.2B by 2027 • Middle Eastern music niche entirely unexplored online	THREATS • Competitors may replicate gaps once identified • Low barriers to entry in online dance market • Technology access disparities in Hunza Valley • Community trust required for data privacy compliance

6. Timeline and Feasibility

The phased approach ensures that technical development is grounded in cultural understanding, system is validated through iterative piloting before main study deployment, and community stakeholders remain engaged throughout the 24-month research process.

Phase	Duration	Title	Key Activities
Phase 1	Months 1–4	Ethnographic Foundation	Cultural interviews, video archive analysis, movement vocabulary documentation, ethics approval
Phase 2	Months 3–8	Technical Development	MediaPipe integration, Unity environment, affective computing module, 3D Blender assets
Phase 3	Months 7–10	Pilot Testing	10 participants, system refinement, movement validation, clinical protocol finalisation
Phase 4	Months 10–18	Main Study	Recruitment, 12-week AKS intervention, clinical assessments, 8-week follow-up
Phase 5	Months 16–24	Analysis & Dissemination	Data analysis, thesis write-up, community feedback, journal submission

7. Market and Commercial Context

7.1 Industry Overview

The online dance education market, valued at approximately **\$3.8 billion in 2022** and projected to reach **\$5.2 billion by 2027**, has expanded rapidly through platforms such as STEEZY (\$20/month), CLI Studios (\$16/month), Dancio (\$25/month), and DancePlug. These platforms primarily offer pre-recorded Western genre content with critical limitations: no real-time feedback, no mental health integration, and a narrow cultural repertoire excluding South and Central Asian traditions.

DanReality's competitive entry point: \$12/month — the most affordable option in the market — combined with unique storytelling integration, live sessions, middle eastern music, and clinically-validated DMT protocols that no competitor currently offers.

7.2 Competitive Advantages

- Only platform integrating Hunza traditional dance with AI-driven DMT biofeedback
- Middle Eastern and Shina/Burushaski music: entirely unexplored niche in online dance
- Live sessions with expert feedback — addressing key gap identified in competitor analysis
- Storytelling integration embedded in dance classes — no competitor offers this
- Twin synchronization as signature pedagogical and therapeutic modality
- Price-competitive entry at \$12/month vs. competitors at \$16–\$25/month

7.3 Target Markets

Primary: Adults 25–55, English/Urdu/Hindi speakers across India, Pakistan, UK, Canada, Germany, Singapore, Australia, Kuwait, Ukraine, and Italy.

Secondary: Youth 16–24 through university and community partnerships; seniors 56+ through wellness programs.

Clinical: Mental health practitioners, DMT therapists, and wellness organizations seeking culturally-integrated dance therapy tools.

8. Expected Contributions

8.1 Theoretical Contributions

This research advances theory at the intersection of **embodied cognition, cultural heritage studies, and affective computing**. The concept of "**Affective Kinematic Synchronization**"—the coupling of emotional expression, movement quality, and interpersonal coordination as a measurable therapeutic mechanism—offers a novel theoretical framework for technology-enhanced DMT in underrepresented cultural contexts.

8.2 Practical Contributions

The AKS prototype delivers a **functional, open-source platform** for culturally-rooted dance therapy adaptable for other traditional dance forms. Technical documentation and 3D cultural assets will be made available under Creative Commons licenses. The system prioritizes **low-cost hardware requirements** (standard webcam, mid-range computer), addressing accessibility barriers in low-resource settings like Hunza Valley.

8.3 Societal Contributions

By centring South and Central Asian embodied heritage within mental health innovation, this research challenges the **Eurocentric bias** prevalent in both DMT practice and technology design. The project creates pathways for youth in Pakistan to engage with mental wellness tools that honour their cultural identity. The digital preservation component ensures that Hunza dance knowledge—transmitted through the JanBroz lineage from grandfather to twin grandsons—remains accessible to future generations.

Vision: DanReality does not merely teach dance. It activates the therapeutic power of cultural memory, twin-bond synchrony, and AI-enhanced movement science to heal, connect, and preserve — all at once.

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